



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATION/ NATIONAL SENIOR CERTIFICATE EXAMINATION

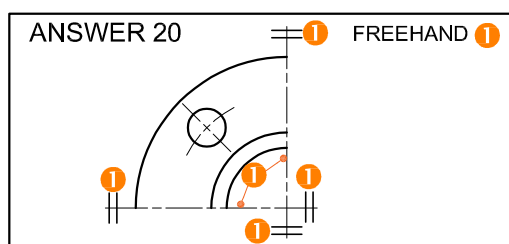
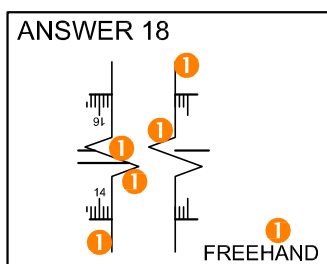
ENGINEERING GRAPHICS AND DESIGN P2

2023

MARKING GUIDELINES

These marking guidelines consist of 7 pages.

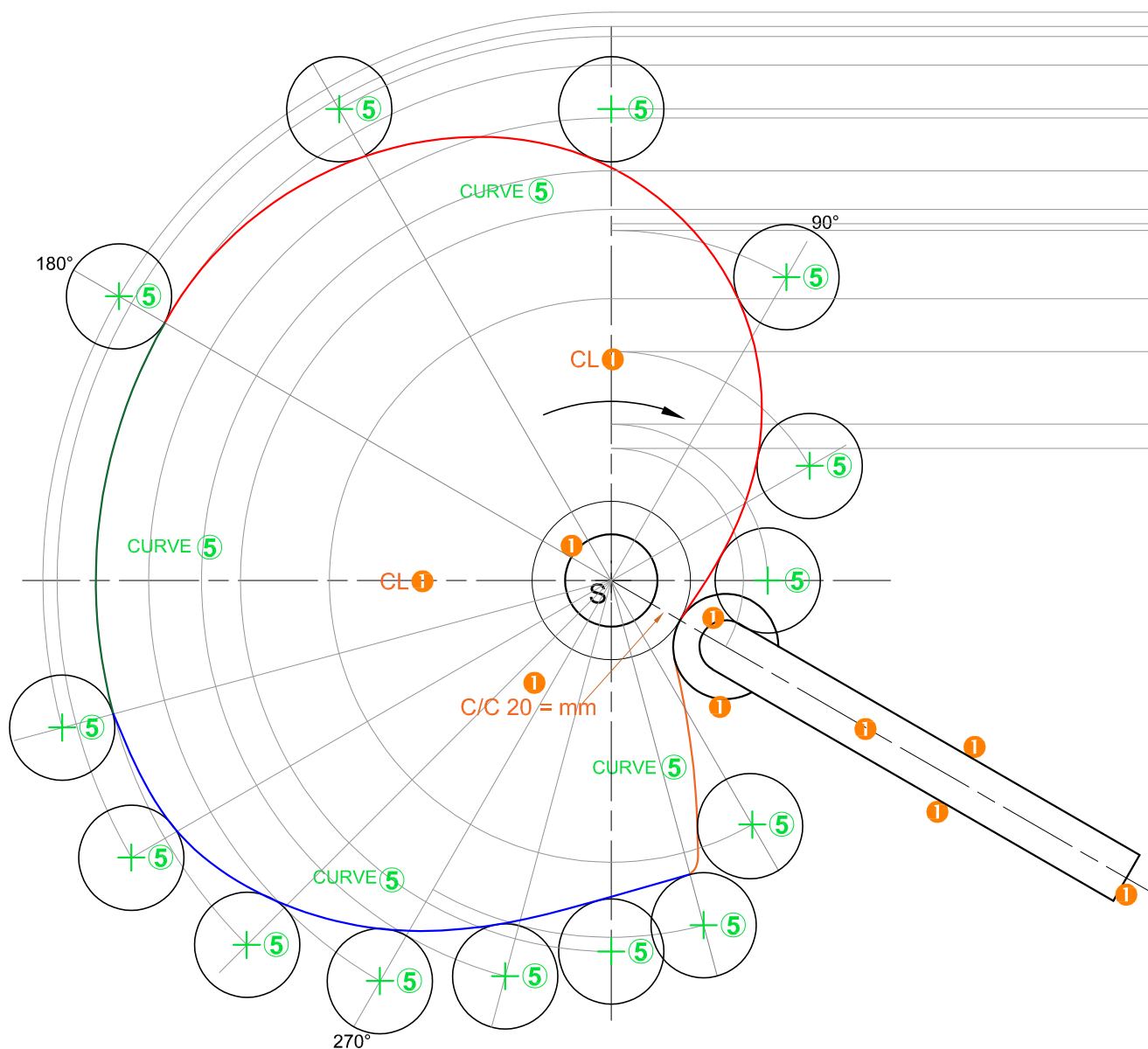
ANSWERS		
1	LISA	1
2	DRAWING NUMBER	1
3	COMBINATION SQUARE	1
4	PRECISION TOOLS	1
5	20 °C	1
6	GLASS	1
7	3000	1
8	THIRD ANGLE	1
9	SECTIONAL RIGHT VIEW	1
10	MULTI PLANE / OFF SET	1
11	REVOLVED	1
12	REMOVED	1
13	C: 17 D: 111 E: 14	3
14	9	1
15	DETAIL T	1
16	KNURLING	1
17	103,025 (MAX.) 102,95 (MIN.)	2
18	See below	3
19	L / =	1
	G / N7	1
	J / HONE	1
	H / 2	1
20	See below	3
TOTAL		30



PAPER 2 QUESTION 1
GRADE 12
JUNE 2023
MARKING GUIDELINE

5. CAM PROFILE

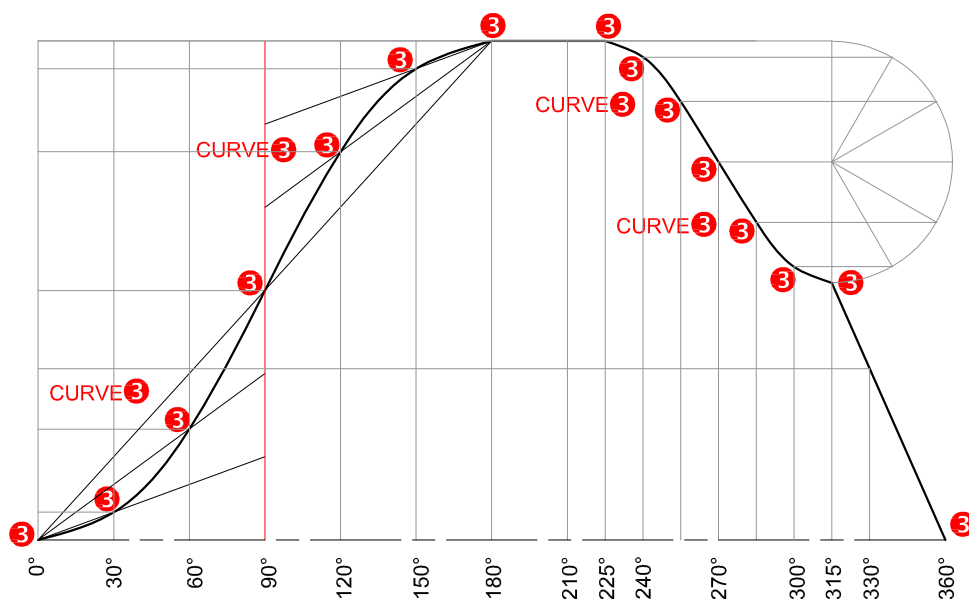
- 14 CENTRE POINTS	7
- ROLLERS	2
- POINTS OF TANGENCY	2
- CURVE TANGENT TO GIVEN FOLLOWER	0,5
- CURVE QUALITY ($4 \times \frac{1}{2}$)	2
TOTAL	$13\frac{1}{2}$

**4. CAM CONSTRUCTION**

- PROJECTION / TRANSFERRING FROM GRAPH	1
- CIRCLE DIVISION	1
- ARC PROJECTION / TRANSFER	1
- PLOT DIRECTION (ANTI-CLOCKWISE)	1
- ARROW (CLOCKWISE)	1
TOTAL	5

ASSESSMENT CRITERIA		
1	GIVEN + MINIMUM DISTANCE + CL	5
4	CAM CONSTRUCTION + ARROW	5
5	PLOTTING + CAM PROFILE	$13\frac{1}{2}$
TOTAL		$23\frac{1}{2}$

PAPER 2 QUESTION 2
GRADE 12
JUNE 2023
MARKING GUIDELINE



DISPLACEMENT GRAPH ②

2. GRAPH CONSTRUCTION

- LENGTH (120) 1
- 12 EQUAL DIVISION 1
- TOTAL 2

2. UNIFORM ACCELERATION + RETARDATION CONSTRUCTION

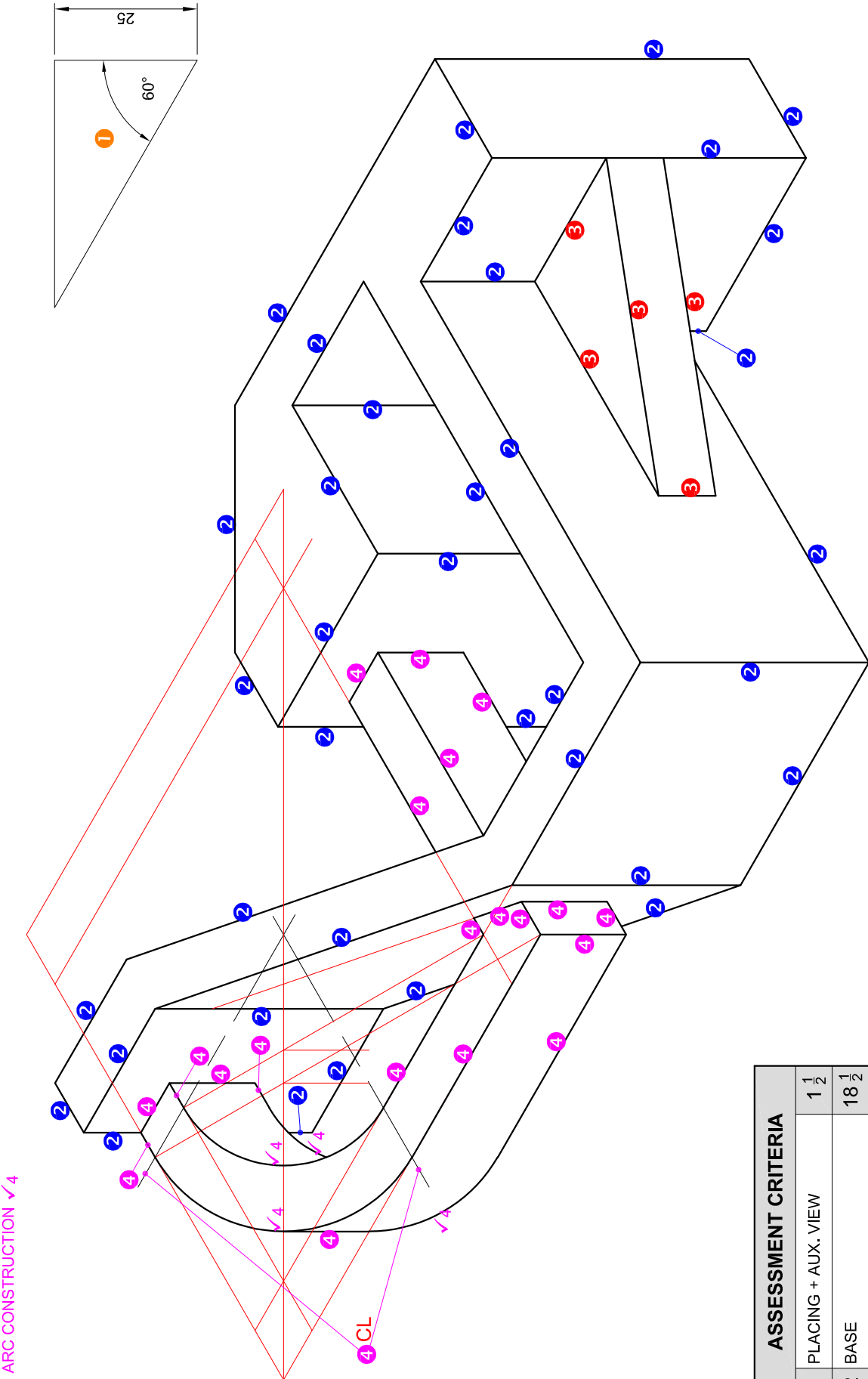
- VERTICAL DIVISION 0.5
- 1st HALF 0.5
- 2nd HALF 0.5
- TOTAL $1\frac{1}{2}$

2. SIMPLE HARMONIC MOTION CONSTRUCTION

- SEMI CIRCLE 0.5
- CIRCLE DIVISION 0.5
- ADDITIONAL 15° DIVISIONS 0.5
- PROJECTION 0.5
- TOTAL 2

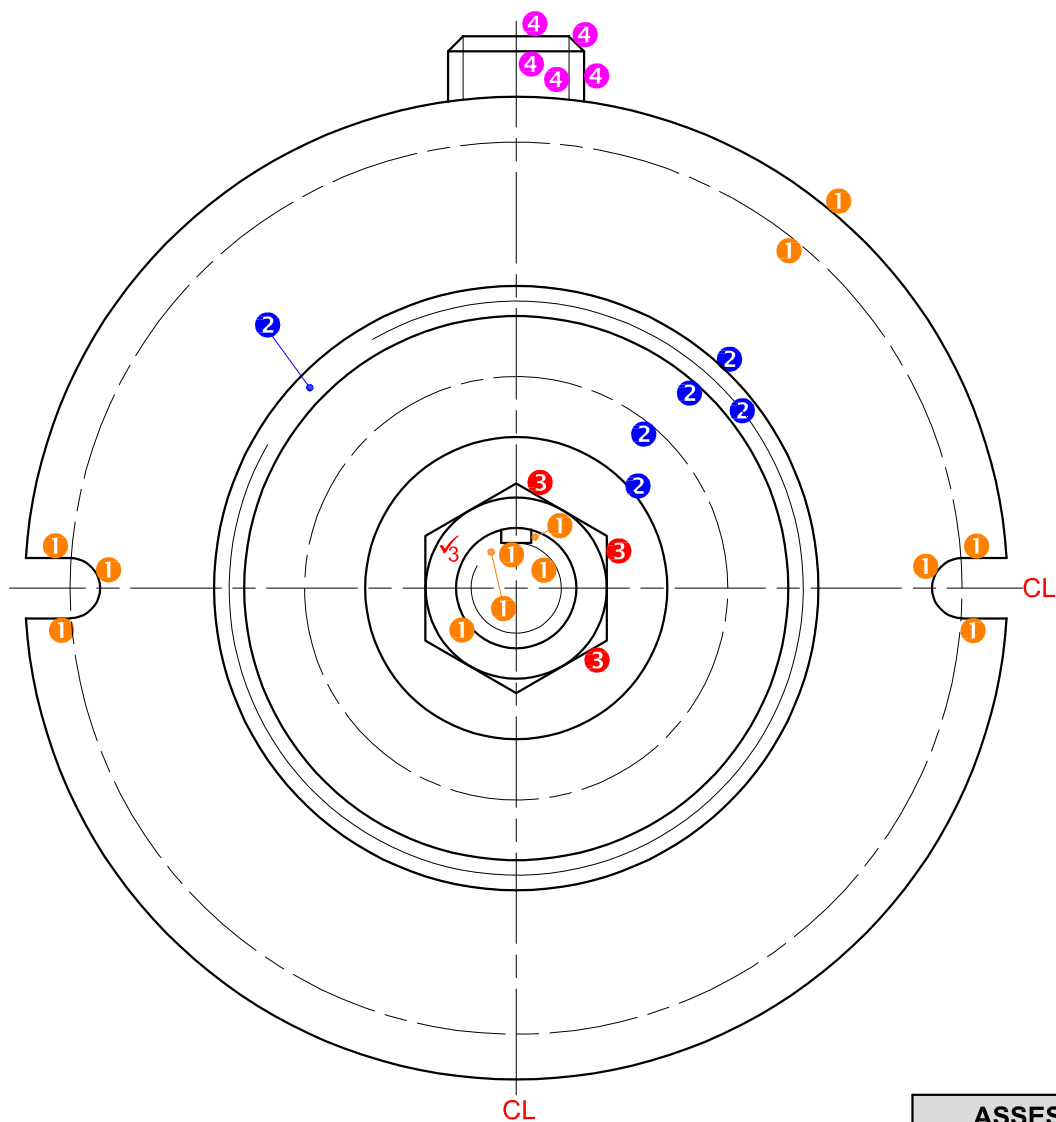
ASSESSMENT CRITERIA		
2	GRAPH CONSTRUCTION + LABELS	6
3	PLOTTING GRAPH + GRAPH CURVES	$9\frac{1}{2}$
TOTAL		$15\frac{1}{2}$

PAPER 2 QUESTION 2
 GRADE 12
 JUNE 2023
 MARKING GUIDELINE



ASSESSMENT CRITERIA	
1	1 1/2
2	18 1/2
3	2 1/2
4	15 1/2
TOTAL	
38	

PAPER 2 QUESTION 3
GRADE 12
JUNE 2023
MARKING GUIDELINE



1. CENTRE LINES
SANS COMPLIANT
1 - 3 = 1; 4 - 5 = 2

1
2
3

NOTE: THE CONVENTION
AND PCD CLs NOT INCLUDED

2. ASSEMBLY

1 MARK FOR EVERY COMPONENT CORRECTLY
ASSEMBLED (9 PARTS - 1) = 8

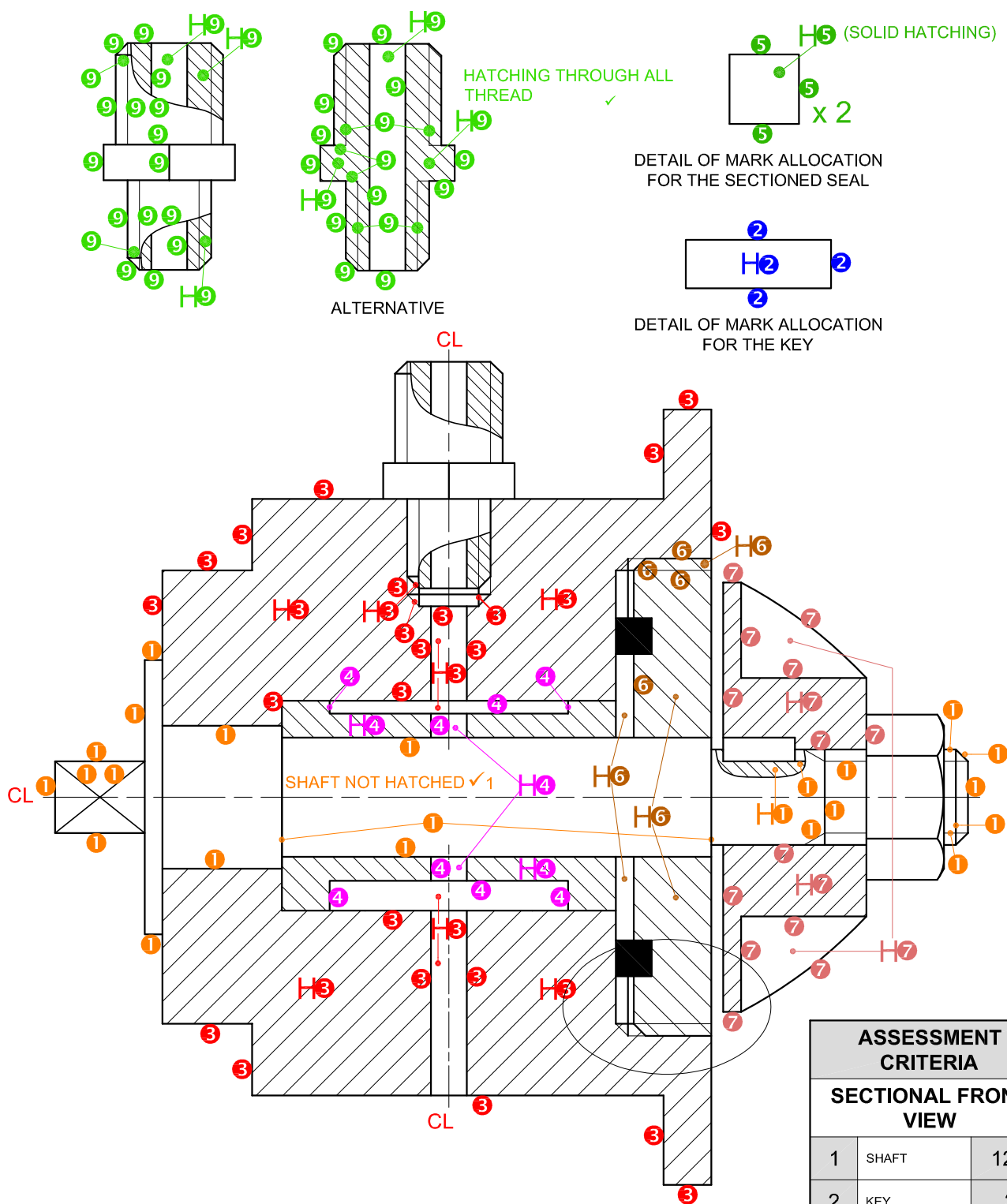
ASSESSMENT CRITERIA		
GENERAL		
1	CENTRE LINES	3
2	ASSEMBLY	8
SUBTOTAL		11

ASSESSMENT CRITERIA

RIGHT VIEW

1	HOUSING + SHAFT	6 $\frac{1}{2}$
2	BACK PLATE + COMPRESSOR WHEEL	3
3	M16 NUT	2 $\frac{1}{2}$
4	OIL LINE FITTING	2 $\frac{1}{2}$
SUBTOTAL		14 $\frac{1}{2}$

PAPER 2 QUESTION 4
GRADE 12
2023
MARKING GUIDELINES



ASSESSMENT CRITERIA		
SECTIONAL FRONT VIEW		
1	SHAFT	12 $\frac{1}{2}$
2	KEY	2
3	HOUSING	18
4	BUSH	5 $\frac{1}{2}$
5	RUBBER SEAL	4
6	BACK PLATE	3 $\frac{1}{2}$
7	COMPRESSOR WHEEL	8
8	M16 NUT	4
9	OIL LINE FITTING	10
SUBTOTAL		67 $\frac{1}{2}$

PAPER 2 QUESTION 4
GRADE 12
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MARKING GUIDELINES